
Later, Lines: Characterizing SAE Feature Manifold Recovery

Camilo Brown-Pinilla

Harvard College

camilobrownpinilla@college.harvard.edu

Abstract

Sparse Autoencoders (SAEs) operationalize the Linear Representation Hypothesis by decomposing neural activations into sums of feature directions. However, evidence increasingly indicates that features in large models often lie on continuous manifolds, and recent work has shown that SAE architecture choice imposes a strong inductive bias on which features are recoverable. We ask: how faithfully do SAEs recover feature manifolds, and does this depend on architecture? Training ReLU+L1, TopK, and JumpReLU SAEs on toy feature manifolds (circle, helix, sphere, torus), we analyze post-activations using two complementary tools: Mapper graphs to assess topological recovery, and kNN graph Laplacian spectra to assess geometric distortion. We find that (i) PCA visualizations can suggest faithful recovery even when topology has been catastrophically destroyed; (ii) ReLU consistently outperforms TopK and JumpReLU in both topological and geometric fidelity, with the latter two architectures frequently shattering connected manifolds into disjoint components despite achieving comparable or lower reconstruction error; and (iii) recoverability is sensitive to sparsity, with even topologically-trivial manifolds like the open helix never recovered across the architectures and sparsity regimes considered. These results suggest that standard PCA-based SAE evaluations hide significant failures of representation, and motivate the development of geometry-aware interpretability methods.

1 Introduction

Mechanistic Interpretability seeks to disentangle the activations of neural networks into atomic, human-interpretable units (known as “concepts” or “features”) to ensure safety and enable control Elhage et al. [2022]. To this end, the Linear Representation Hypothesis (LRH) Park et al. [2023] has emerged as a popular and widely accepted conjecture for how concepts are organized. Roughly speaking, the LRH posits that concepts exist as simple linear directions in the residual stream of large networks. Indeed, several concepts have been found to be represented linearly, including sentiment Tigges et al. [2023], truthfulness Marks and Tegmark [2024], spatial and temporal coordinates Gurnee and Tegmark [2024], and refusal Arditi et al. [2024].

Drawing inspiration from sparse coding and dictionary learning, Sparse Auto Encoders (SAEs) have emerged as a class of simple networks that directly operationalize the LRH. By reconstructing neural activations subject to sparsity constraints, SAEs learn to decompose activations into a sparse sum of feature directions (also referred to as “atoms”).

However, despite the popularity and empirical backing of the LRH, recent research has identified features that are decidedly non-linear, including circles for days of the week and months of the year Engels et al. [2024], and helices for numerical magnitudes Levy and Geva [2024]. From the various feature geometries that have been found, it becomes increasingly evident that the LRH may be better thought of as a special case of a more general *manifold* representation hypothesis. Concurrently

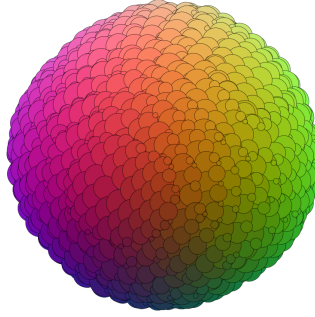


Figure 1: Mapper visualization of the sphere. Nodes are colored by spherical harmonics.

to these findings, it has been shown that the choice of SAE architecture is actually a fundamental *inductive bias*, defining the types of features that SAEs can detect Hindupur et al. [2025].

Together with this finding, the fact that SAEs are designed to pick out feature *directions* even though many features live on *shapes* raises a natural question: **can SAEs recover feature manifolds, and is this ability architecture dependent?** In this paper, we provide evidence that the answer to these questions might be *yes*. Concretely we study SAEs trained on toy manifolds from a novel topological and geometric perspective and show:

- Architecture choice predictably determines recoverability
- Within architectures, recoverability increases as sparsity increases
- Evidence of manifold *shattering*, where connected components are shattered into disjoint pieces *despite clean PCA projections and low MSE*.

We hope that this work will serve as a step towards geometry-aware interpretability and motivate further use of topological and geometric tools for the rigorous study of neural geometry.

2 Background

2.1 SAEs

Roughly speaking, SAEs are a neural parameterization of the sparse coding problem Olshausen and Field [1996], which seeks to represent data in $\mathbb{R}^d$ as sparse codes in $\mathbb{R}^m$, $m \gg d$ where each non-zero element denotes the presence of an atom (feature) from dictionary $D \in \mathbb{R}^{d \times m}$. More formally, let $x \in \mathbb{R}^d$ be a data point, $z \in \mathbb{R}^m$ be a sparse code, and $D \in \mathbb{R}^{d \times m}$ be a dictionary. SAEs solve the following optimization problem:

$$\arg \min_{z, D} \|x - Dz\|_2^2 + \lambda \mathcal{R}(z),$$

$$z = \sigma(Wx + b), \quad W \in \mathbb{R}^{m \times d}$$

where $\mathcal{R}$ is a sparsity promoting regularizer with scalar hyperparameter λ and σ is an element-wise nonlinearity. In practice, the dictionary D and encoder W are often tied, i.e., $D = W^T$.

This formulation describes the standard family of single-layer SAEs and includes all architectures we consider; some recent variants (e.g., Gated SAEs Rajamanoharan et al. [2024a], Matryoshka SAEs) extend it with additional structure but are out of scope for this work. Defining an architecture amounts to choosing a nonlinearity σ and regularizer $\mathcal{R}$. In this work, we use the ReLU+L1 Bricken et al. [2023], TopKMakhzani and Frey [2013], and JumpReLU Rajamanoharan et al. [2024b] architectures.

2.2 Topological Data Analysis

Topological Data Analysis (TDA) seeks to recover information about the topology of point cloud data, roughly providing some sense of the “shape” of a dataset. TDA has been applied across diverse domains, including computational biology Nicolau et al. [2011], neuroscience Giusti et al. [2015],

materials science, and machine learning, where it has been used to study loss landscapes Horoi et al. [2022] and the internal representations of neural networks Naitzat et al. [2020].

Persistent homology Edelsbrunner et al. [2002], Carlsson [2009] and the Mapper algorithm Singh et al. [2007] stand out as the most prevalent TDA tools. Given point cloud data, persistent homology computes topological invariants by covering points with open balls of varying radius, recording the size of the different homology groups H_n (these are known as the Betti Numbers β_n , which capture the number of n -dimensional “holes” in a space) as a function of radius. While persistent homology can recover rich topological data, it is often unstable in practice and scales poorly with the number of data points, with computation past H_1 for large point clouds typically being intractable. We elect not to use it for these reasons.

Mapper Mapper is an algorithm that, in addition to recording the size of the first two homology groups, produces informative visuals representing the “shape” of a dataset. Intuitively, Mapper can produce the 1-skeleton (i.e., graph) of a dataset. More precisely, given a dataset $\mathcal{X} \subset \mathbb{R}^d$, Mapper first works by projecting the data via a filter function $f : \mathbb{R}^d \rightarrow \mathbb{R}^p$, where p is typically at most 3 or 4. The image $f(\mathcal{X})$ is then covered with overlapping open cubes. Following the covering, the points in the preimage of each cube are clustered, with each cluster becoming a node in the graph. Finally, edges are drawn between nodes that exist in overlapping regions of the open cover.

Like the varying radii in persistent homology, the size of the open cover and the amount of overlap between open cubes are hyperparameters that allow topological information to be extracted across scales. In our experiments, we establish a hyperparameter grid and use the convention that the Betti pair (β_0, β_1) belonging to the largest stable region of the grid is the true shape, and that the other Betti pairs are noise. This convention has the added benefit of allowing us to have a sense of how noisy or brittle the topology of the point clouds we study is. Unlike persistent homology, Mapper is typically cheap to compute because of the reduction in dimensionality via the filter f . In practice, one typically chooses a dimensionality reduction technique like PCA or UMAP. We use the projection onto the first four eigenvectors of the Laplacian of the data’s kNN graph, which we motivate in the following section.

2.3 The Laplace Beltrami Operator

The Laplace-Beltrami Operator $\Delta_{\mathcal{M}}$ is the generalization of the familiar Laplacian Δ for functions on $\mathbb{R}^n$ to functions defined on Riemannian manifolds $\mathcal{M}$. In particular, the spectrum (λ_i, u_i) of eigenvalues λ_i and eigenfunctions u_i characterize the *intrinsic geometry* of a manifold $\mathcal{M}$. In fact, the eigenfunctions themselves are intrinsic coordinates: for the unit circle S^1 , the first non-trivial eigenfunctions are $(\cos \theta, \sin \theta)$, isometrically embedding S^1 into $\mathbb{R}^2$; for the flat torus T^2 , the first four are $(\cos \theta, \sin \theta, \cos \phi, \sin \phi)$; for S^2 , the first three are degree-1 spherical harmonics, equal to the Cartesian coordinates restricted to the sphere.

Connection to kNN Graphs Given samples $\{x_i\}_{i=1}^N$ from $\mathcal{M} \subset \mathbb{R}^D$, the Coifman–Lafon graph Laplacian

$$L = I - \tilde{D}^{-1}\tilde{W}, \quad \tilde{W} = D^{-1}WD^{-1}, \quad W_{ij} = \exp(-\|x_i - x_j\|^2/2\sigma^2)$$

built from Gaussian edge weights on a kNN graph and density-corrected via the $\alpha = 1$ normalization $\tilde{W}$, converges to $\Delta_{\mathcal{M}}$ as $N \rightarrow \infty$ and $\sigma \rightarrow 0$ at appropriate joint rates Coifman and Lafon [2006], Belkin and Niyogi [2008]. Convergence holds for both eigenvalues and eigenvectors, so the first k non-trivial eigenvectors of L approximate the first k Laplace-Beltrami eigenfunctions of $\mathcal{M}$, providing a discrete estimate of the manifold’s intrinsic coordinates that we can compute without knowing $\mathcal{M}$ *a priori*.

Importantly, because we work with toy manifolds with well-known spectra, the discrepancy between the discrete spectrum and true spectrum tells us precisely how much our recovered points $\mathcal{Z} = \{z_i\}_{i=1}^N$ are distorted with respect to the original feature manifold, giving us local, geometric information in addition to the global, topological perspective that Mapper gives us. Indeed, this duality motivates our choice of Mapper filter function as projection onto the Laplacian eigenvectors as this is the “true” filter in the sense that each data point is mapped to its position in the intrinsic coordinate system of the underlying manifold, ensuring that the Mapper cover respects the data’s geometry rather than depending on an embedding-specific projection like PCA. Indeed, Figure 1 was produced with this filter function, precisely recovering the sphere.



Figure 2: Mapper graph of torus recovery by ReLU(a) and TopK(b) at the same sparsity level. TopK distorts geometry and has much lower topological stability than ReLU.

3 Methods

3.1 What Does it Mean to Recover a Manifold?

When considering whether an SAE recovers a feature manifold, one must decide *where* to look for recovery: the preactivations $Wx + b$, the postactivations z , or the dictionary D . Certainly, studying the preactivations is noninformative, since affine maps preserve topology and the power of the architecture lies in nonlinearity. Between the dictionary and post-activations, we argue that the post-activations z are the correct choice. To see why, consider an SAE trained on a circular feature. The SAE can represent this faithfully with a basis span or approximate it with evenly spaced atoms around the circle, defining a spectrum from representation to memorization. The circle can certainly be reconstructed from any of these configurations, with the basis span being the most accurate and efficient representation.

Suppose then that we train an SAE with an atom count of 4 that represents the circle perfectly as a basis span. Immediately, we see the issue with studying the dictionary: despite perfect representational capacity, the dictionary is topologically 4 distinct points, making circular structure invisible to any topological tool. On the other hand, for reasonable sample size N , the activations $\{z_i\}^N$ tile the circle and can be pushed arbitrarily close together, making the correct circular topology easily recoverable. Indeed the dictionary elements can be viewed as the 1-skeleton of the activations, making the $\{z_i\}^N$ the clear choice of object to study for combined topological and geometric analysis.

3.2 Topological Recovery via Mapper

As mentioned in the background, Mapper provides us with both a faithful visualization of the data shape using the natural Laplacian filter, and also provides us with the first two Betti numbers via the Euler formula $\chi(G) = \beta_0 - \beta_1 = V - E$. Recall again that by varying the resolution and overlap of the cover, Mapper also gives a sense of topological recoverability across scale. Using a shared hyperparameter grid, we can then define the *topological stability* $\tau \in [0, 1]$ as the percentage of configurations in the grid that produced the correct Betti pair.

3.3 Geometric Recovery via Spectral Fidelity

Conditional on topological recovery, we can consider some notion of geometric distortion from the underlying feature manifold. Without topological recovery, such a comparison would not be sensible in the same way that it does not make sense to speak of how deformed a collection of disjoint points is compared to a circle. As mentioned before, the theoretical connection between the Coifman-Lafon spectrum and the Laplace-Beltrami spectrum lets us define such a notion in terms of the eigenvalue discrepancy. In particular, we can consolidate this information into a single scalar we call the *spectral*

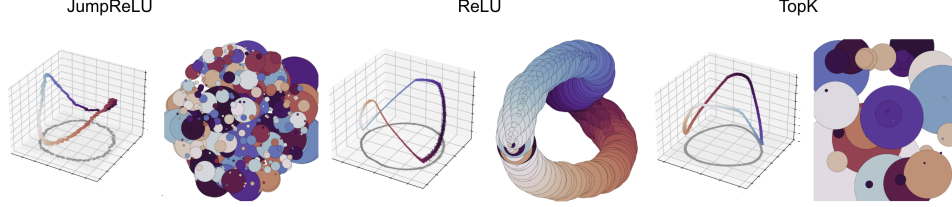


Figure 3: PCA plots vs. Mapper graphs for JumpReLU, ReLU, and TopK on the circle. All SAEs trained at the same sparsity. Despite faithful looking PCA plots for every architecture, only ReLU preserves circle topology. Both JumpReLU and TopK demonstrate *shattering*.

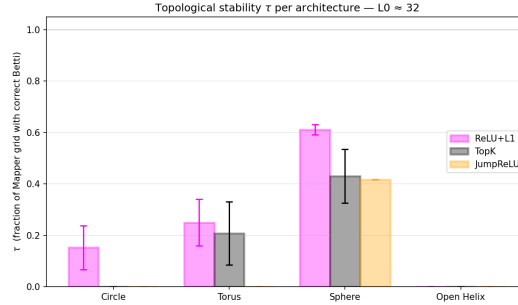


Figure 4: Topological stability τ across architectures and manifolds ($\ell_0 \approx 32$).

error:

$$\mathcal{E} = \frac{1}{K} \sum_{i=2}^K |\log \lambda_i / \lambda_1 - \log \lambda_i^* / \lambda_1^*|$$

Where the λ_j^* refer to the true eigenvalues and the log is applied to compress the range and prevent large eigenvalues from dominating the score. Normalization by the first eigenvalue is applied to ensure values are on the same scale. In particular, note that this encodes our conditional assumption: λ_1 is nonzero if and only if the recovered manifold is connected. When the manifold is shattered and topology is lost, $\lambda_1 = 0$ and the metric is undefined, returning ∞ . We say that $\mathcal{E} < 0.05$ is an "almost isometric recovery" since the metric on the true, noisily sampled manifolds returned a value in a small neighborhood of 0.025.

3.4 Experimental Setup

We evaluate ReLU, TopK, and JumpReLU on four manifolds: the helix, the circle, the sphere, and the torus. All manifolds are embedded in $\mathbb{R}^{64}$ via their canonical parameterizations. All SAEs have latent dimension 128 and are trained on $\approx 12M$ samples drawn uniformly from the manifold. After training, activations $\mathcal{Z} = \{z_i\}^N$ are computed from $N = 100,000$ samples x_i again drawn uniformly from each surface. Each SAE is trained for 50,000 steps. We calibrate hyperparameters on each manifold to target atom counts $K \in \{8, 16, 24, 32\}$. Notably, we were unable to find an ℓ_1 coefficient that reliably produced an atom count of 16 on the circle for ReLU. Due to the effect degenerate points had on the eigensolver, analysis of lower atom count SAEs was mostly intractable. However, some runs did complete and we report them here. We run each (architecture, sparsity, manifold) with 3 different seeds to control for the effects of random initialization.

4 Results

PCA Plots are Not All You Need Literature engaging with SAEs trained on non-linear features often uses PCA plots of activations as confirmation that the concept manifold is properly represented. In Figure 3, we plot the PCA projection of different architectures trained on the circle next to their mapper graphs. Each architecture has approximately the same atom count, giving them the

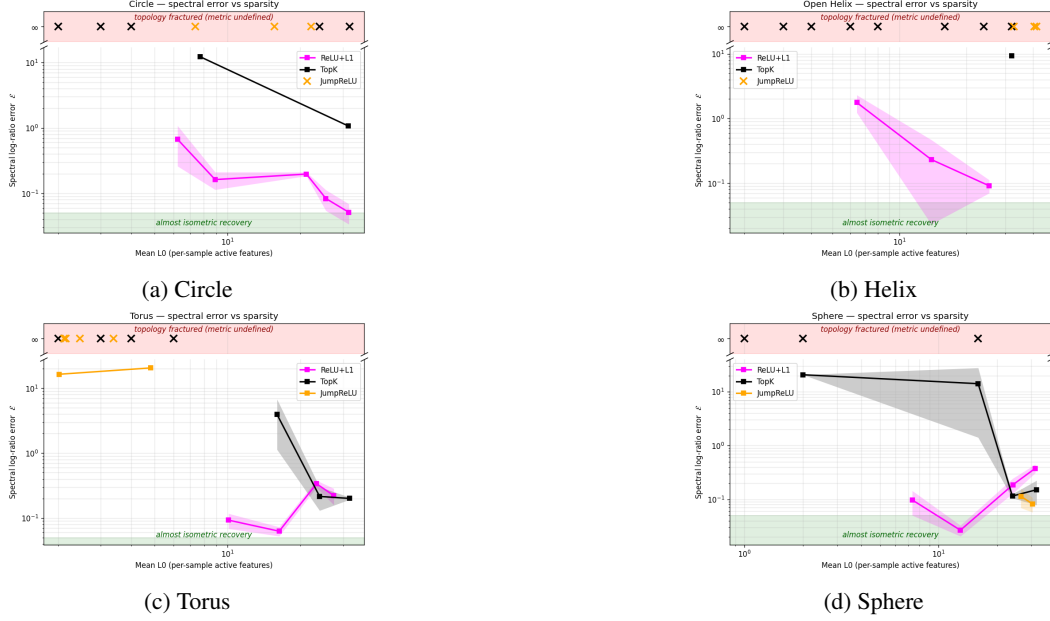


Figure 5: Spectral Error $\mathcal{E}$ vs. sparsity (atom count) across manifolds

same representational capacity. Visually, PCA plots would imply that each architecture does indeed preserve the circle. However, topological analysis reveals a stark contrast. The mapper graphs for both JumpReLU and TopK reveal significant shattering of the circle’s single connected component ($\beta_0 = 1$) into several hundred disjoint points, indicating complete destruction of the original manifold. This trend is consistent across the manifolds considered.

Notably, ReLU perfectly preserves the circle up to homotopy, with $(\beta_0, \beta_1) = (1, 1)$ revealing a single connected component and single loop. Indeed, the spectral error $\mathcal{E} = 0.193$ further confirms topological recovery ($\mathcal{E} < \infty$) but points to nontrivial geometric distortion, serving as additional evidence that PCA does not reveal accurate topological or geometric information. As the “cherry on top”, we note that the mean MSE of TopK and JumpReLU was only ≈ 0.001 compared to ≈ 0.0001 for ReLU, hiding the failure of the latent representation behind good reconstruction.

Recoverability is Architecture Dependent In Figure 5, we display the spectral error of each architecture on each manifold as a function of sparsity. Note that $\mathcal{E}$ is plotted on a log scale. Of particular note is that ReLU is consistently better than both other architectures both in terms of spectral error and robustness. By robustness, we mean that ReLU produces only finite values (implying connectivity is preserved) while both TopK and JumpReLU consistently fracture with both finite and infinite error values appearing at the same sparsity level indicating that recovery is partially determined by random initialization.

Figure 4 further confirms these observation. ReLU consistently outperforms TopK and JumpReLU in terms of topological stability, meaning that the correct topology is visible across a wider range of Mapper configurations. By using Mapper together with Spectral analysis, we can consider both the global topological perspective and local geometric perspective and find that the results agree in both cases.

Trouble for 1-Manifolds One quite notable result is that in both the topological and geometric sense, recovery of the 1-manifolds we consider (the circle and helix) is more difficult. This is quite counterintuitive, as these are the simplest spaces we consider. In particular, the helix is *never topologically recovered*. While geometrically more complex than the circle, the open helix is homeomorphic to an open interval and therefore contractible (homotopy-equivalent to a single point) making its expected Betti numbers $(\beta_0, \beta_1) = (1, 0)$ the simplest non-trivial topological signature. This is particularly concerning that real features have been found to be organized on helices across

models [Levy and Geva, 2024, Engels et al., 2024]. These results point to the need to scrutinize current SAE methodology when applied to seemingly simple feature manifolds.

5 Discussion

By studying SAEs on toy feature manifolds, we are able to introduce a novel topological and geometric perspective to the study of feature manifolds. More specifically, by taking advantage of the theoretical link between the spectra of kNN graphs and the Laplace-Beltrami operator, we are able to consider notions of geometric recovery in a principled manner. By using topological tools, we are able to confirm the spectral results and further classify failure modes in terms of shattering.

The results we see are quite surprising, and point to limitations in the current study of feature manifolds. Most notably, we show the surprising amount of crucial information that PCA hides given the convincing visuals it produces. With PCA serving as a canonical interpretation tool in the current literature, our results open the door to question whether this information is truly the full story and if more informative tools and analysis paradigms can be developed.

Finally, the stark contrast between ReLU and the TopK and JumpReLU architectures serves to confirm previous research establishing architecture choice as a highly nontrivial consideration. As we move steadily away from the linear representation hypothesis towards more general notions of feature organization, we hope that this work will serve as motivation for development of geometry-aware interpretability methods.

Future Work The results and approaches presented in this work generate several exciting directions for future work. Of course, one can consider extending this analysis past the toy setting to real model features. Using tools like persistent homology, one can characterize the topology of model features and confirm whether circular features are truly circular or of this is yet another PCA artifact. Further, while there is no ground-truth comparison for real features, one can perform the same spectral analysis introduced here using the spectrum of the kNN graph built from model activations as a baseline.

Along the lines of graphs, a subtle point is that the graph (or simplicial complex) of feature co-activations is intimately related to activation topology in the sense that the coactivation graph may be the 1-skeleton of activations. While this has not been proved, we posit this as a conjecture. Notably, should this be true, this would enable extremely cheap topological analysis of SAEs.

Finally, we comment on potential reasons for *why* the architecture results organize themselves this way. We posit that manifold recoverability is intrinsically tied to the "niceness" of the SAE architecture. More precisely, ReLU SAEs are globally Lipschitz, while TopK introduces discontinuity through its global top-k selection and JumpReLU through its threshold gating. We conjecture that the discontinuity of TopK and JumpReLU is what permits the manifold-shattering we observe. We conjecture that SAEs architectures can be further formalized in terms of their functional properties in such a way that would reveal rich theoretical insights as to why exactly we observe the results presented here.

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